COMP5310: Principles of Data Science

W6: Hypothesis Testing and Evaluation

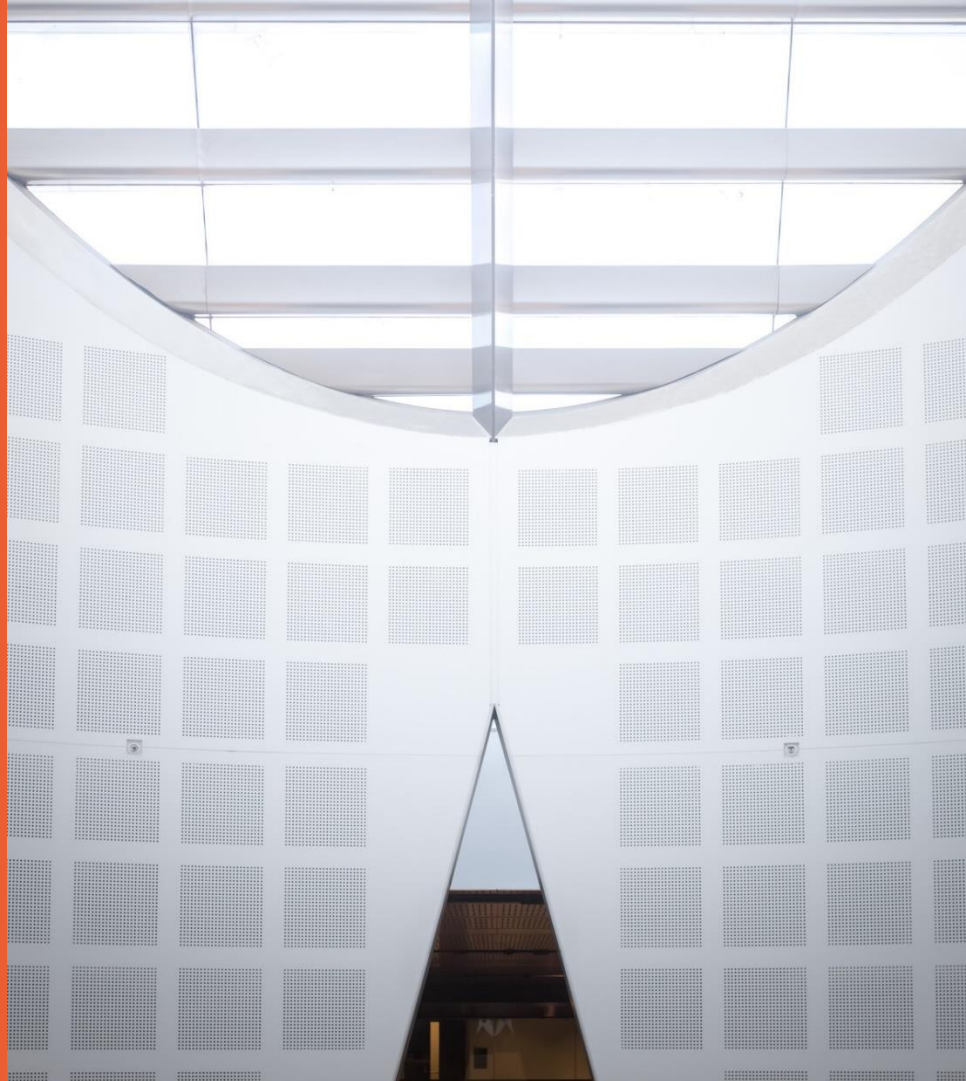
Presented by

Dr. Ying Zhou

School of Computer Science

Based on slides by previous lecturers of this unit of study

THE UNIVERSITY OF SYDNEY



Last week: Querying and summarising data

Objective

- To be able to extract a data set from a database, as well as to leverage on the SQL capabilities for in-database data summarisation and analysis.

Lecture

- Data gathering reprise.
- SQL querying.
- Summarising data with SQL.
- Statistic functions support in SQL.

Readings

- Data Science from Scratch: Ch 24.

Exercises

- Data Loading.
- SQL Querying.
- Python DB Querying.
- Data Summarization using SQL.

TO-DO in W5

- Finish Ed Lessons Python modules.
- Finish Ed Lessons SQL modules.

Goal of today's lecture

- High-level overview of statistical tests (not a deep dive)
- Understand how to compare groups and evaluate differences using data
- Learn how to choose appropriate statistical tests for common data scenarios

From EDA to Hypothesis Testing



What we did in EDA

- Explore data using
  - Summary statistics (mean, median, standard deviation, etc)
  - Visualisations (histograms, box plots)
- Identify patterns and differences
- E.g.
  - Employees in large organization seems spent less time working from home than those in smaller organization.
  - Industry A seems support working from home more than industry B

Employee number	Percentage working from home	Total number of respondents
Between 1 and 4	0.443956044	91
Between 20 and 199	0.287664042	381
Between 5 and 19	0.331896552	233
I am the only employee	0.636507937	63
More than 200	0.255893099	739

The limitation of EDA

- Are these differences real or just random variation?
- Would we see the same pattern with a different sample?
  - E.g. what if we run the same survey but with different people?
- EDA gives insight but not certainty
- What we need next is *a way to test whether observed differences are statistically meaningful*
  - This helps us compare groups, evaluation difference and make data driven decisions

STATISTICAL SIGNIFICANCE TESTING

THE UNIVERSITY OF SYDNEY

The Research Question

- In data analysis, we often ask:
  - Are two groups **different**?
  - Is one approach **better than another**?
  - Is an observed pattern **real**, or just **random variation**?
- Example
  - In our data set, we observe that average WFH percentage in industry A is different to that in industry B
- But the data set is only a sample
  - Often we want to know if the difference exists in the population (e.g all individuals in industry A and industry B), or
  - It just happens in the those responded to our data set.

Hypothesis testing

- We use hypothesis testing to specify whether to accept or reject a claim about a population depending on the evidence provided by a sample of data.
- A hypothesis test examines two opposing hypotheses about a population:
  - The **null hypothesis** and the **alternative hypothesis**.
  - The null hypothesis (H_0) represents the assumption that there is no real difference between groups, or relationship between features.
  - The alternative hypothesis is the statement that the different/relationship exists or in certain direction.
- **Example:** Determine whether the mean of a population differs significantly from a specific value or from the mean of another population.

Testing reliability with p-values

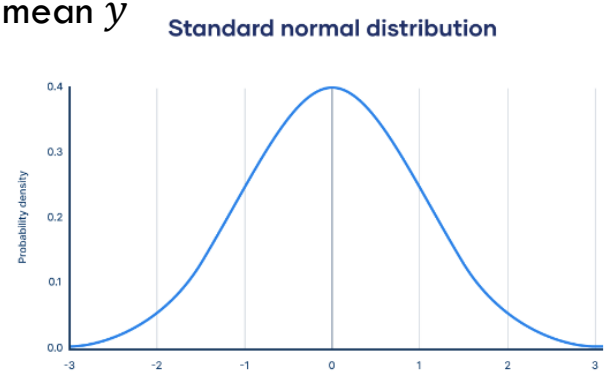
- Most tests calculate a **p-value** for measuring observation extremity, that is, the probability of observing data at least as extreme as what we saw, assuming H_0 is true
- Compare to significance level threshold α.
  - α is the probability of (wrongly) rejecting H_0 given that it is true (Type I error rate, i.e., false positive).
  - Commonly use α of 5% or 1%.

Actual Condition \ Test results	Accept H_0	Reject H_0
H_0 (H_0 is True) - No difference	Right Decision	Type I error
H_1 (H_0 is False) - Difference exists	Type II error	Right Decision

P-value	Indicates	Reject H_0?
$<\alpha$	Strong evidence against the null hypothesis	Yes
$>\alpha$	Weak evidence against the null hypothesis	No
$=\alpha$	Marginal	NA

General idea

- Suppose we have two normal random variables X and Y with the same known variance, we want to test whether they have the same mean
  - Sample 100 random numbers from X, and calculate its mean $\bar{x}$
  - Sample 100 random numbers from Y, and calculate its mean $\bar{y}$
  - Let μ_x and μ_y be the mean of X and Y, respectively
  - How do we conclude whether $\mu_x = \mu_y$ from the value of $\bar{x} - \bar{y}$



- **null hypothesis:** $\mu_x = \mu_y$
- **alternative hypothesis:** $\mu_x \neq \mu_y$
- P-value: probability of generating two sets of samples of 100 each such that the difference between their empirical means is at least $|\bar{x} - \bar{y}|$, under the assumption that the null hypothesis is true

One-side test

- Suppose we want to test whether $\mu_x > \mu_y$
  - **null hypothesis:** $\mu_x = \mu_y$
  - **alternative hypothesis:** $\mu_x > \mu_y$
  - P-value: probability of obtaining two samples **result at least as extreme as the observed result (e.g.** $\bar{x} - \bar{y}$**, in the direction of the alternative hypothesis,** assuming H_0 is true.
- P-value (in general):
  - Two sided:
    - P(observed or more extreme outcome on both directions | H0 true)
  - One side:
    - P(observed or more extreme outcome on one direction| H0 true)

Not every test result is correct

- α=0.05 will erroneously reject H_0 5% of the time
- Perform enough tests and you will get a false result (p-hacking)
- Good science:
  - Determine hypotheses before looking at data
  - Perform hypothesis-agnostic data cleaning
  - Remember that p-values do not replace common sense
- https://sites.uw.edu/stlab/2016/03/09/the-arbitrary-magic-of-p-0-05/

Choosing a statistical test

- There are many statistical tests
- No single test works for all situations
- The choice of often depends on
  - What you want to compare
  - Type and shape of data
  - How the data was collected

What determines the choice

- Number of grups
  - 2 groups vs. 3 or more groups
- Relationship between samples
  - Independent groups
  - Paired/same subjects
- Date type/distribution
  - Numerical
  - Ordinal (e.g. Likert scale)
  - Normal vs. skewed

Do we always need statistical tests?

- Not always
- Statistical tests are used when we only have a sample and we want to use that to infer about a population
- But sometimes we do have the population
- Example:
  - An e-commerce company records **all sales transactions**
  - Compare:
    - This quarter vs last year's same quarter

TYPES OF STATISTICAL STUDIES

THE UNIVERSITY OF SYDNEY

Types of statistical studies

Observational Study

- Simply observing what happens.
- Records information about subjects without applying any treatments to subjects (passive participation of researcher).
- Most data sets we have used so are are from observational study
  - E.g. the work from home survey data, the water measurement data

Experimental Study

- Records information about subjects while applying treatments to subjects and controlling study conditions to some degree (active participation of researcher).

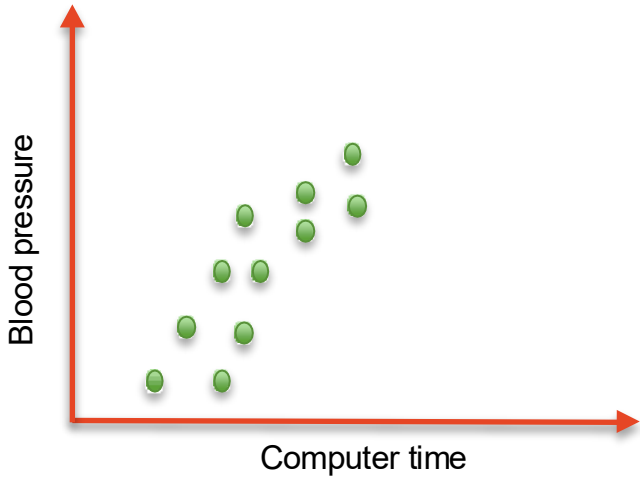
Observational studies

Sample survey

- Provide information about a population based on a sample at a specific point in time.
- Only establish correlation not causality!
  - **Study 1:** Tanning and Skin Cancer.
    - The observational study involved 1,500 people.
    - Selected a group of people who had skin cancer and another group of people who did not have skin cancer.
    - Asked all participants whether they used tanning beds.
    - Wanted to see if there was an association between tanning beds and skin cancer prevalence.

Observational studies

- **Study 2:** Average Computer Time vs Blood Pressure.
  - Enrol 100 individuals in the observational study.
  - Ask them about the average computer time they spent each day.
  - Measure their blood pressure.



Experimental studies

- Strong hypotheses, sample size for desired power and controlled data collection per specified protocols.
- Establish causality.
- **Example:** randomized control trials.
  - 100 subjects.
  - Factor: Average Computer Time.
  - Treatments:
    - Control group (computer time: max. 30 minutes).
    - Treatment group (computer time: 2 hours).
  - 50 subjects randomly assigned to each treatment.
  - Response: we measure the blood pressure for each group.

Experimental vs observational

- Main difference between observational studies and experiments
  - Most experiments use random assignment while observational studies do not.
- Observational studies typically only establish correlation but not causality
- Experimental studies establish causality

Types of variables

Dependent variable

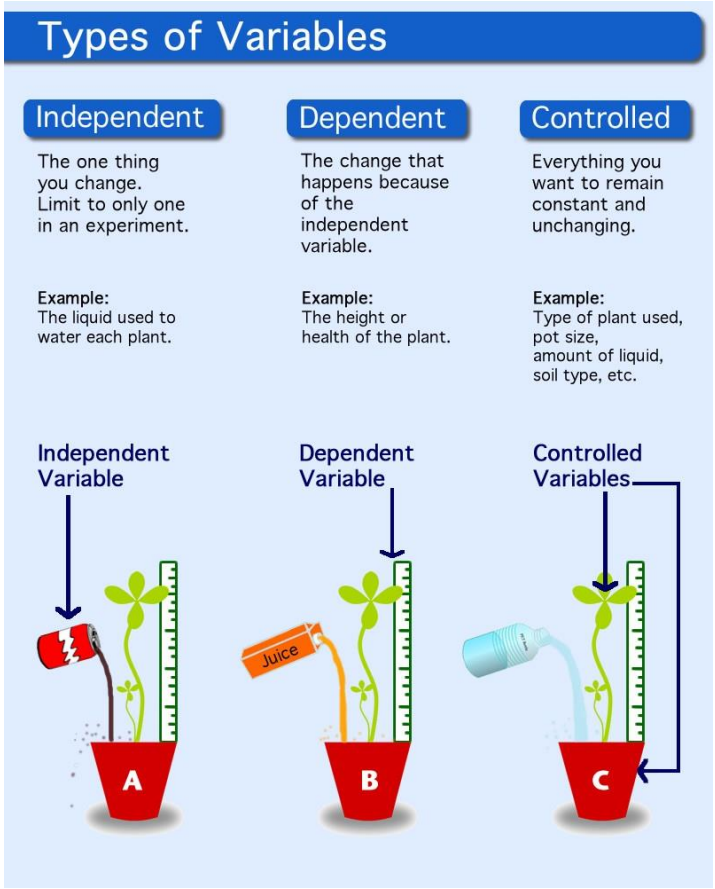
– Measure of interest.

Independent variable

– Manipulated to observe the effect on dependent variable

Controlled variables

– Materials, measurements and methods that don't change.



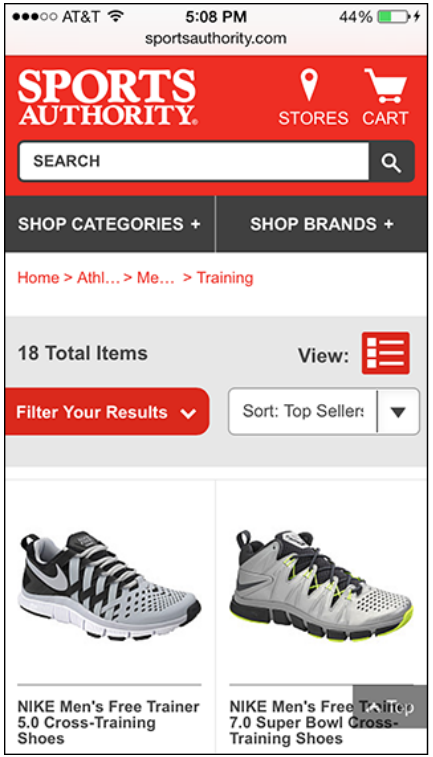
http://edtech2.boisestate.edu/angelacovil/506/procedure.html

TESTING WHICH APPROACH IS BETTER BETWEEN SUBJECTS

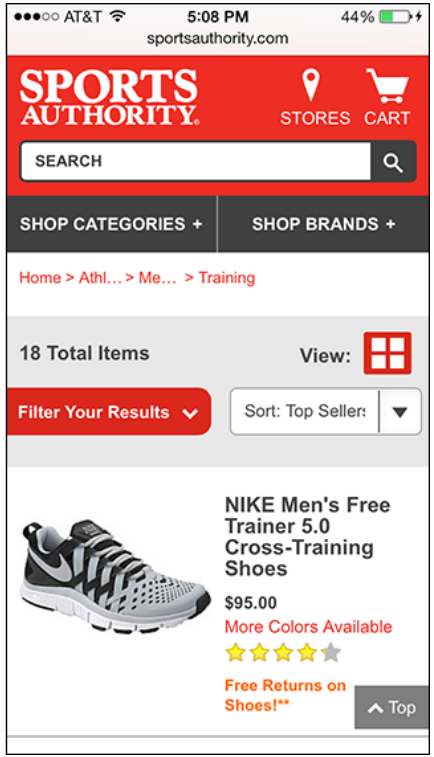
THE UNIVERSITY OF SYDNEY

Scenario: Comparing visual layouts

Grid view



List view



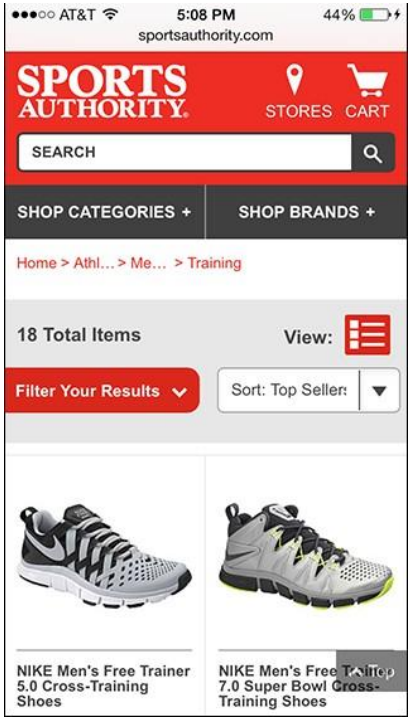
https://www.nngroup.com/articles/image-vs-list-mobile-navigation/

Research question

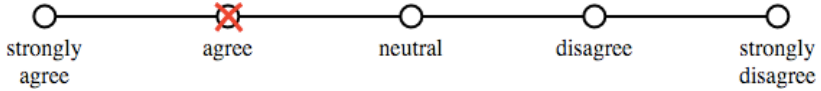
Do users prefer grid view?

Data/Measurement: User ratings of layouts

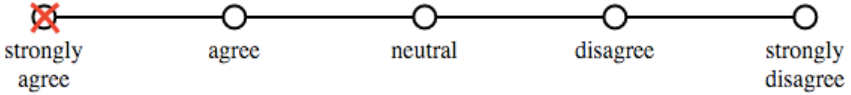
Example response from User Group A



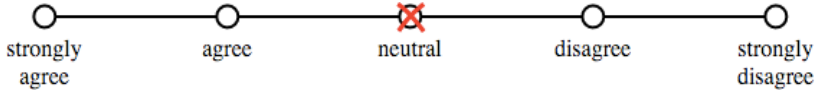
Page is easy to use.



Page gives good overview.

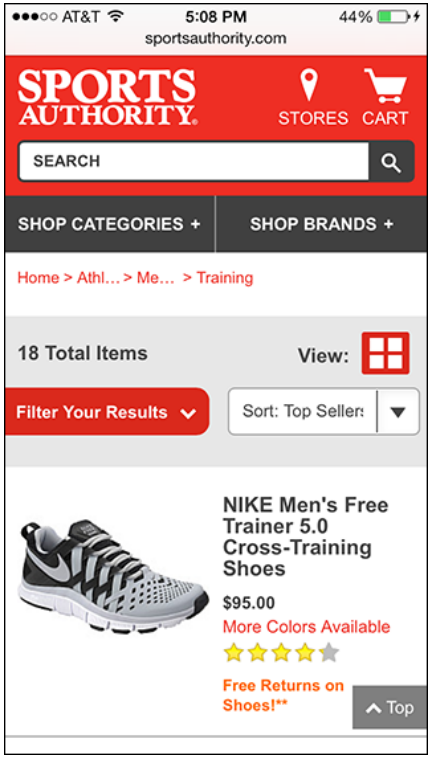


Page gives sufficient detail.

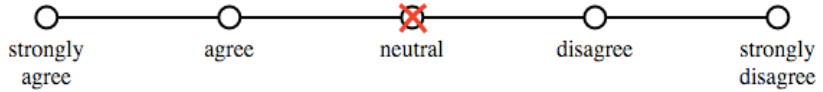


Data/Measurement: User ratings of layouts

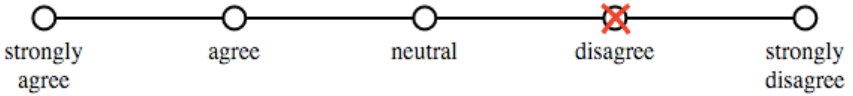
Example response from User Group B



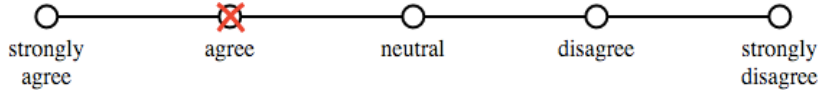
Page is easy to use.



Page gives good overview.



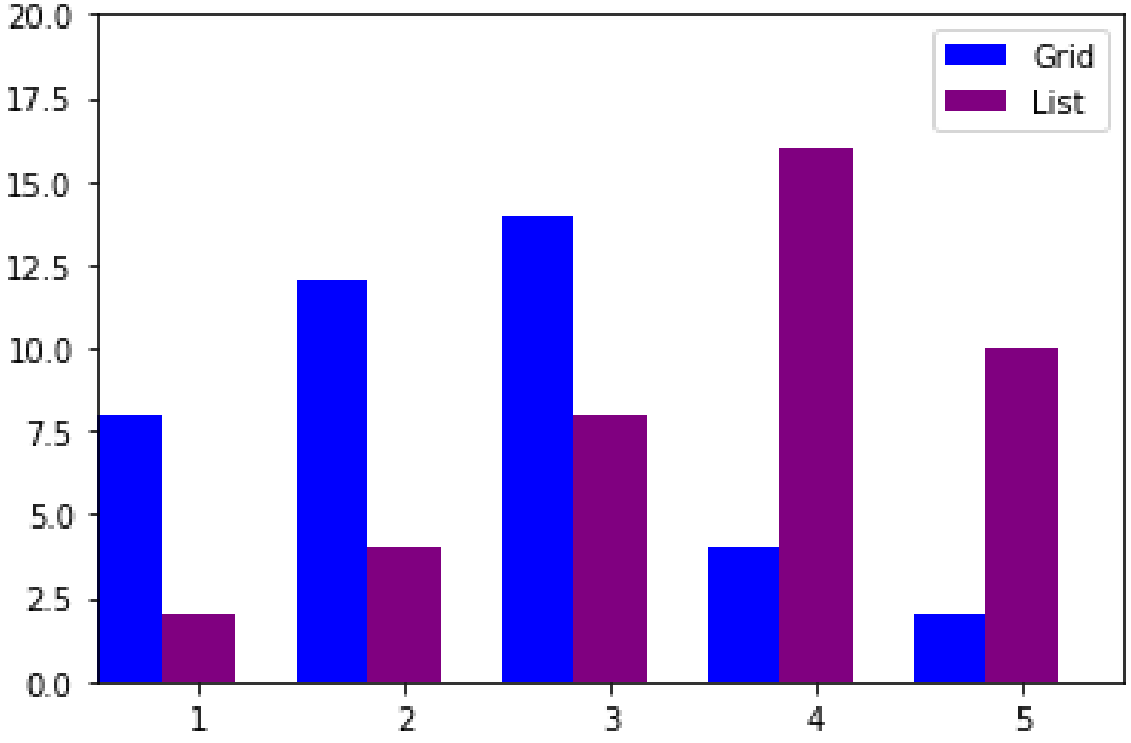
Page gives sufficient detail.



Generate ratings data

- We assume different subject groups for each condition.
- Each subject sees one of the layouts and is asked to rate on a 5-point Likert scale how strongly he agree or disagree with the statement:
- Question to subjects: Page gives a good overview?
- 1=strongly agree; 2=agree; 3=neutral; 4=disagree; 5=strongly disagree.
  - **G_data** = [1, 3, 3, 2, 4, 2, 3, 3, 1, 5, 2, 3, 4, 2, 1, 3, 2, 2, 1, 3, 2, 3, 4, 2, 1, 3, 2, 2, 1, 3, 1, 3, 3, 2, 4, 2, 3, 3, 1, 5]
  - **L_data** = [4, 5, 2, 4, 4, 3, 5, 4, 3, 5, 1, 4, 5, 3, 4, 4, 2, 3, 4, 5, 1, 4, 5, 3, 4, 4, 2, 3, 4, 5, 4, 5, 2, 4, 4, 3, 5, 4, 3, 5]
- **G_data** corresponds to ratings from users that see the **grid view**.
- **L_data** corresponds to ratings from users that see the **list view**.

Visualise ratings data



Setup: Comparing two versions of a display

- Subjects are users of the display (or summary, interface, etc).
  - **Dependent variable** is user rating (or comprehension, etc).
  - **Independent variable** is the version of the display.

- **Problem:** Find out which version of a display is better.
- **Question:** Do users prefer Grid view?
- **Null hypothesis (H_0):** there is no difference between Grid view and List view.

Significance: Unpaired Student's t-test

- Tests the null hypothesis that two population means are equal.
- **Assumptions:**
  - The samples are *independent*.
  - Populations are *normally distributed*.
  - Standard deviations are *equal (by default)*.
- https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ttest_ind.html#scipy.stats.ttest_ind

Significance: Mann-Whitney U test

- Nonparametric version of unpaired t-test.
  - Test the null hypothesis that the distribution underlying sample x is the same as the distribution underlying sample y.
- **Assumptions:**
  - The samples are *independent*.
- **Note**
  - N should be at least 20.
- https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.mannwhitneyu.html#scipy.stats.mannwhitneyu

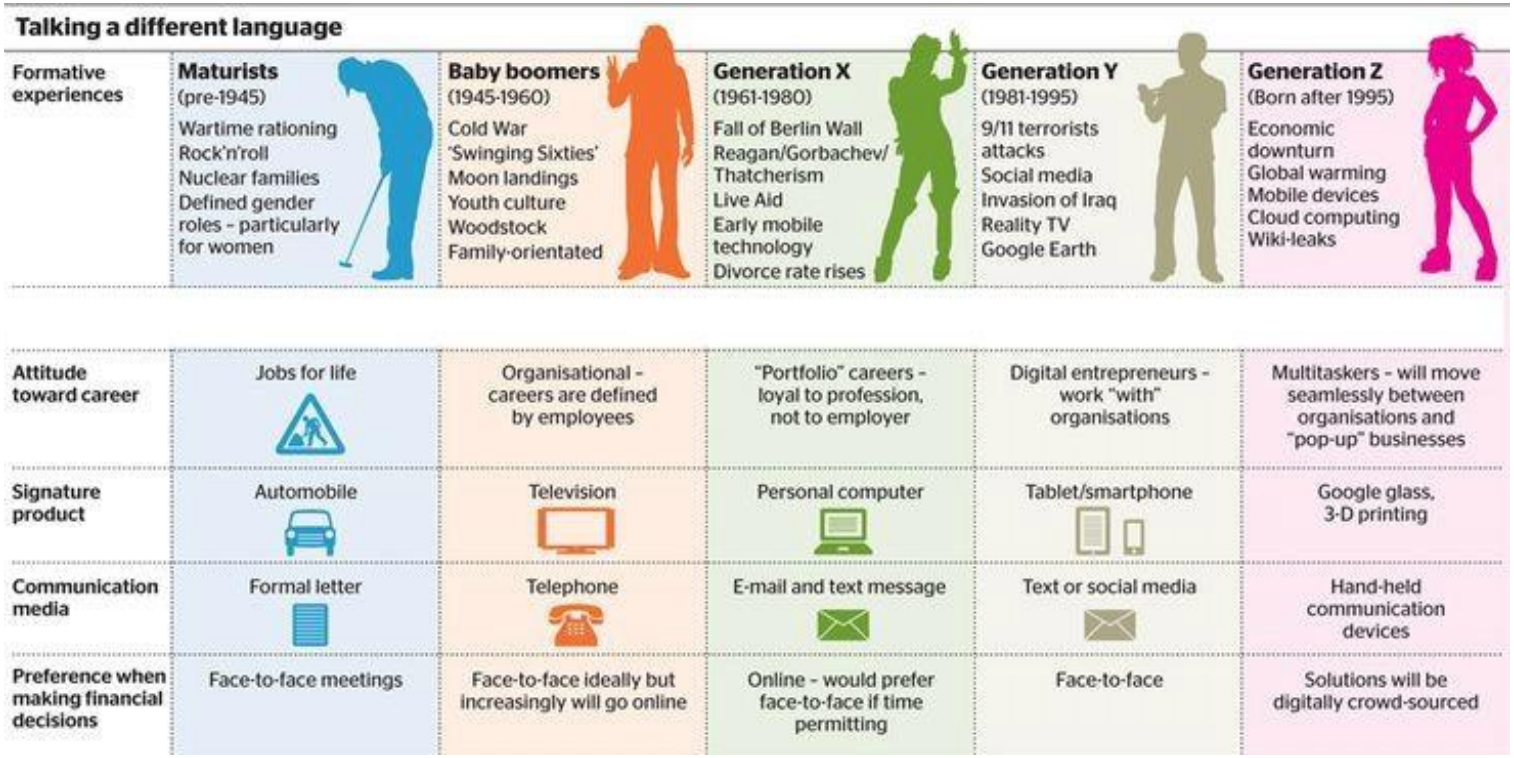
Exercise: Comparing visual layouts

- Test for difference
  - Run the code cell under “Test whether grid view is preferred”
  - Do users prefer grid view?

TESTING WHETHER MULTI-GROUPS DIFFER

THE UNIVERSITY OF SYDNEY

Scenario: Mobile use by generation



https://ihumanmedia.com/2015/09/14/gen-x-millennials-vs-baby-boomer-real-estate-baby-work-travel-politics-shopping/

Research question

Does mobile use differ across generations?

Data/Measurement: Survey of mobile use

- May be collected by survey or user data.
- **Dependent variable:**
  - Number of texts per day.
- **Independent variable:**
  - Generation {B,G,M}.

Texting survey

1. What year were you born?

2. How many texts do you send per day?

Significance: Analysis of variance (ANOVA)

- Tests the null hypothesis two or more groups have the same population mean.
- **Assumptions:**
  - The samples are *independent*.
  - Populations are *normally distributed*.
  - Standard deviations are *equal*.
- https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.f_oneway.html#scipy.stats.f_oneway

Significance: Kruskall-Wallis H-test

- Nonparametric version of ANOVA.
  - Test the null hypothesis that the population median of all of the groups are equal
- **Assumptions:**
  - Samples are *independent*.
- **Note:**
  - Not recommended for samples smaller than 5.
  - Not as statistically powerful as ANOVA.
  - Both ANOVA and Kruskall-Wallis H-test are extensions of the Unpaired Student's t-test and Mann-Whitney test used to compare the means of more than two populations.
- https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.kruskal.html#scipy.stats.kruskal

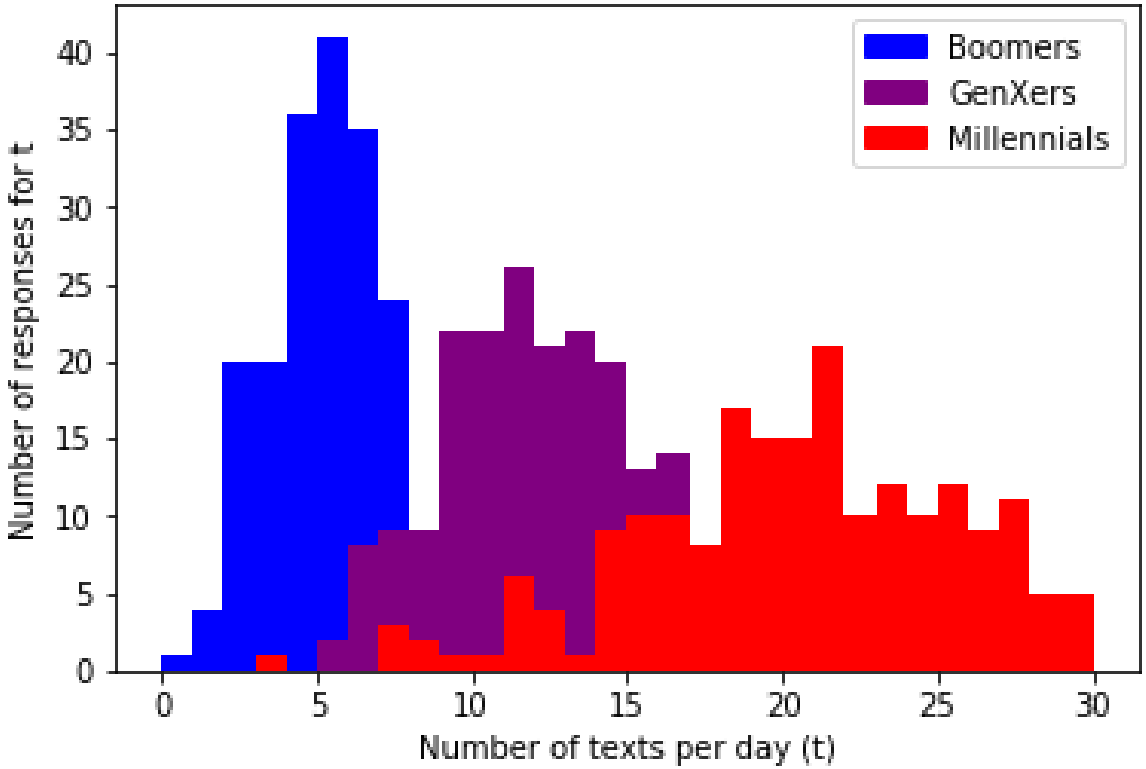
Setup: Comparing behaviour across groups

- Subjects are rows of data.
  - **Dependent variable** is number of texts per day.
  - **Independent variable** is generation {B,G,M}.
- **Q:** Is there any difference between groups?
- **H_0:** Group means (or medians, for nonparametric methods) are the same

Generate generation data

- Imagine we conducted a survey of **200 baby boomers** (born 1945-1960), **200 generation Xers** (born 1961-1980) and **200 millennials** (born 1981-1995).
- For the purposes of this exercise, let's generate some simulated samples. We assume:
  - **Baby Boomers** send 5 texts per day on average with standard deviation 2.
  - **GenXers** send 12 texts per day on average with standard deviation 3.
  - **Millennials** send 20 texts per day on average with standard deviation 5.

Visualise generation data



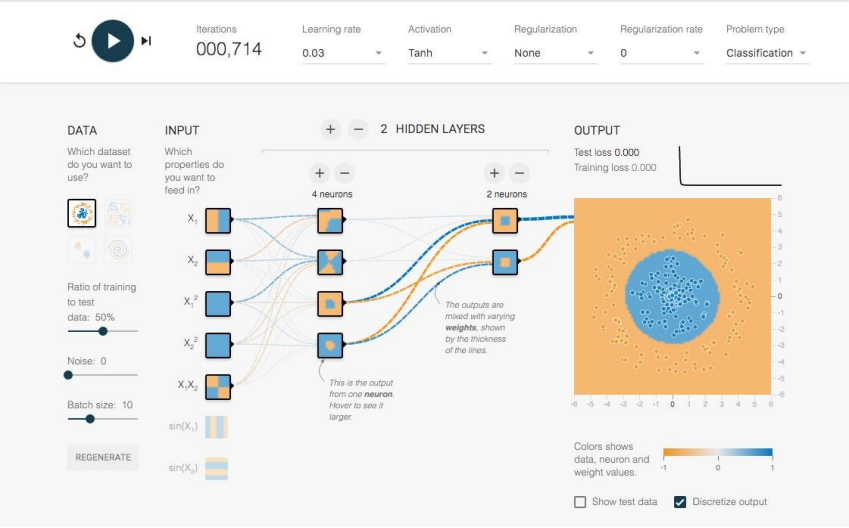
Exercise: Comparing mobile behaviour

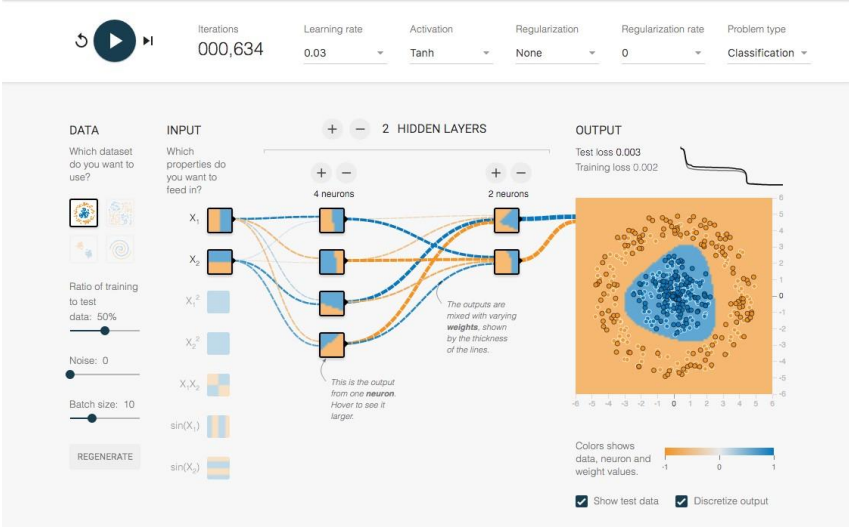
- Test for difference
  - Run the code cell under “Testing for differences”
  - Does the data satisfy ANOVA assumptions?

TESTING WHICH APPROACH IS BETTER WITHIN SUBJECTS

THE UNIVERSITY OF SYDNEY

Example scenario: Comparing classifiers





http://playground.tensorflow.org/

Research question

Does my new model perform better?

Task: Spam/ham detection

- Let's assume our classifiers predict whether an email is:
  - 1: spam.
  - 0: ham.
- In classification, we compare
  - Predicated labels vs true labels
- This gives four outcomes: TP,FP, TN, FN
- All metrics are derived from them

https://scikit-learn.org/stable/modules/model_evaluation.html

Measurement: Accuracy, precision, recall, f1

Model prediction

Actual results	Spam (s=1)	Ham (s=0)
Spam (g=1)	*TP* *(true positives)*	*FN* *(false negatives)*
Ham (g=0)	*FP* *(false positives)*	*TN* *(true negatives)*

- **Accuracy**: percentage of correct over all instances.
  - (TP+TN) / N
- **Precision**: percentage of correct system predictions.
  - TP / (TP+FP)
- **Recall**: percentage of correct gold labels.
  - TP / (TP+FN)
- **F1**: Harmonic mean of Precision and Recall.
  - 2PR / (P+R)

Evaluating classifier: Holdout & cross-validation methods

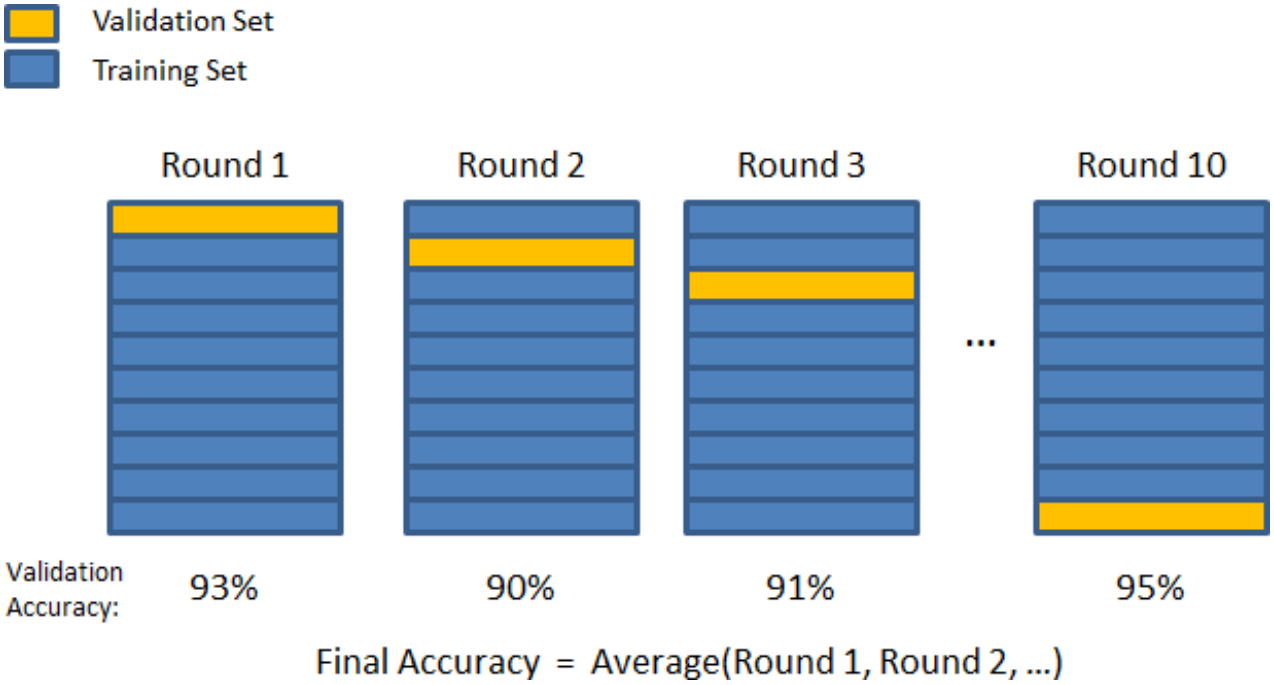
Holdout method

- Splits the data randomly into two independent sets.
  - Training set (e.g., 2/3) for model construction.
  - Test set (e.g., 1/3) for evaluation estimation.
  - Result from one split may not be representative enough

Cross-validation (*k*-fold, where k = 10 is most popular)

- Randomly partition the data into k mutually exclusive subsets, each approximately equal size.
- Repeat k times:
  - Train on k−1 folds
  - Test on remaining fold
- Average performance across all folds
- Leave-One-Out is a particular form of cross-validation:
  - *k* folds where *k* = # of tuples, for small sized data.

Data: Cross validation



https://chrisjmccormick.wordpress.com/2013/07/31/k-fold-cross-validation-with-matlab-code/

Using Hypothesis Testing

- For each fold:
  - Compute accuracy for Model A and B
- Now we have
  - Multiple paired results
- We can run
  - Paired t-test or the non-parametric version

Significance: Paired Student's t-test

- Tests the null hypothesis that two population means are equal.
- **Assumptions:**
  - The samples are *paired* (e.g. before and after a treatment).
  - Populations are *normally distributed*.
  - Standard deviations are *equal*.
- https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ttest_rel.html#scipy.stats.ttest_rel

Significance: Paired tests for non-parametric data

- Nonparametric version of paired t-test.
- **Assumptions:**
  - The samples are *paired*.
- **Note:**
  - Often used for ordinal data, e.g., Likert ratings.
  - N should be large, e.g., ≥20.
- https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.wilcoxon.html#scipy.stats.wilcoxon

Generate gold and classifier labellings

- We generate **10,000 gold labels.**
  - Marking approximately **20% as spam** (1) based on a random number generator and the rest as ham (0).
    - 0: 8000 and 1: 2000
- **System 1** incorrectly marks 5% of ham as spam and fails to detect 20% of actual spam.
- **System 2** incorrectly marks 10% of ham as spam and fails to detect 10% of actual spam.

System 1

		PREDICTED 1	PREDICTED 0
ACTUAL	1	1600 (TP)	400 (FN)
	0	400 (FP)	7600 (TN)

System 2

		PREDICTED 1	PREDICTED 0
ACTUAL	1	1800	200
	0	800	7200

Setup: Comparing classifiers

- Subjects correspond to cross-validation folds.
  - **Dependent variable** is some measure of accuracy (precision, recall, f1, etc).
  - **Independent variable** is the algorithm, feature set, etc.
- **Q**: Is my shiny, new model better?
- $\mathbf{H_0}$: Accuracy is not better for the new model.

Exercise: Comparing models

- Generate data
  - Run the code cell under “Generate gold and classifier labelling”
  - Run the code cell under “Split data into folds”
- Calculate accuracy
  - Run the code cell under “Calculate classifier accuracy”
  - Run the code cell under “Calculate scores across folds”
- Test for differences
  - Run the code cell under “Compute significance for sys1 and sys2”

REVIEW

Additional reading (not examinable)

Some great online resources:

- Hypothesis testing, power, sample sizes
  - https://online.stat.psu.edu/stat415/
- What does it all even mean?
  - https://plato.stanford.edu/entries/statistics/

W6 review: Hypothesis testing and evaluation

Objective

- Overview of hypothesis testing and learn Python tools for hypothesis testing and classifier evaluation.

Lecture

- **Significance**
  - Pairwise t-tests.
  - Non-normal data.
- **Evaluation**
  - Classifier evaluation.
  - User evaluation.

Readings

- Data Science from Scratch: Ch 7.
- Model evaluation

Exercises

- Scipy: statistical tests.
- Sklearn: evaluation metrics.